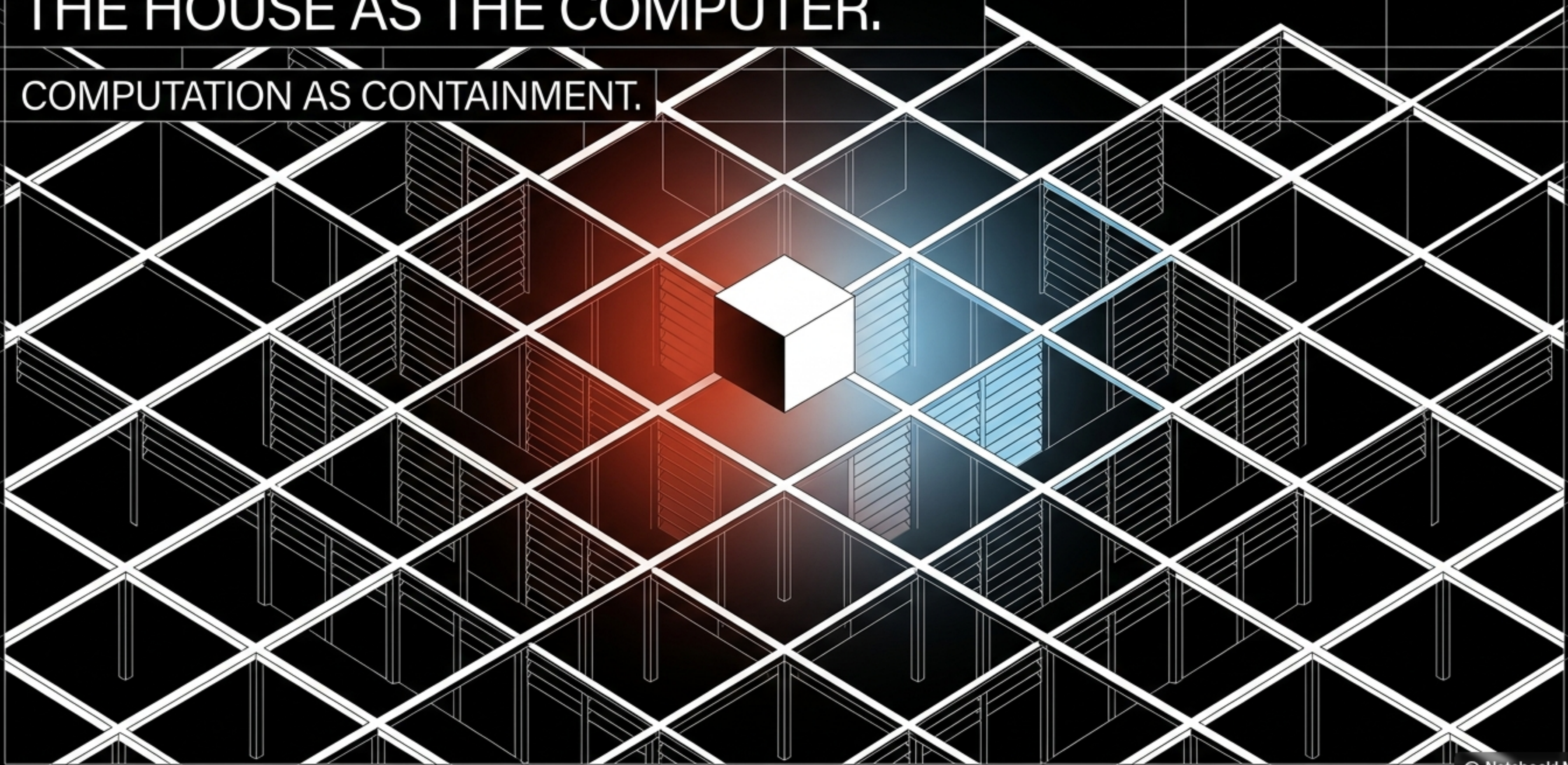


RADIATIVE SWITCH COMPUTATION: THE HOUSE AS THE COMPUTER.

COMPUTATION AS CONTAINMENT.



THE EVOLVING THERMAL GRID

The efficiency story treats dissipation as a resource to capture. Embodiment treats it as the computation itself. Heat is the terminal sink into which usable energy decays. A conventional computer reaches it after informational work. A radiative switch shapes the physical path to that decay so that arriving at it constitutes having computed something.

	THE DOMESTICATED VERSION (DISTRICT HEATING)	RADIATIVE SWITCH COMPUTATION (EMBODIMENT)
SUBSTRATE & HARDWARE	Independent. Silicon routed to water pipes.	The thermal material itself. Waxes and louvers.
REACTION TO ENVIRONMENT	Computes identically whether the room is 10°C or 30°C.	Evolving physical thermal dynamics are the computation.

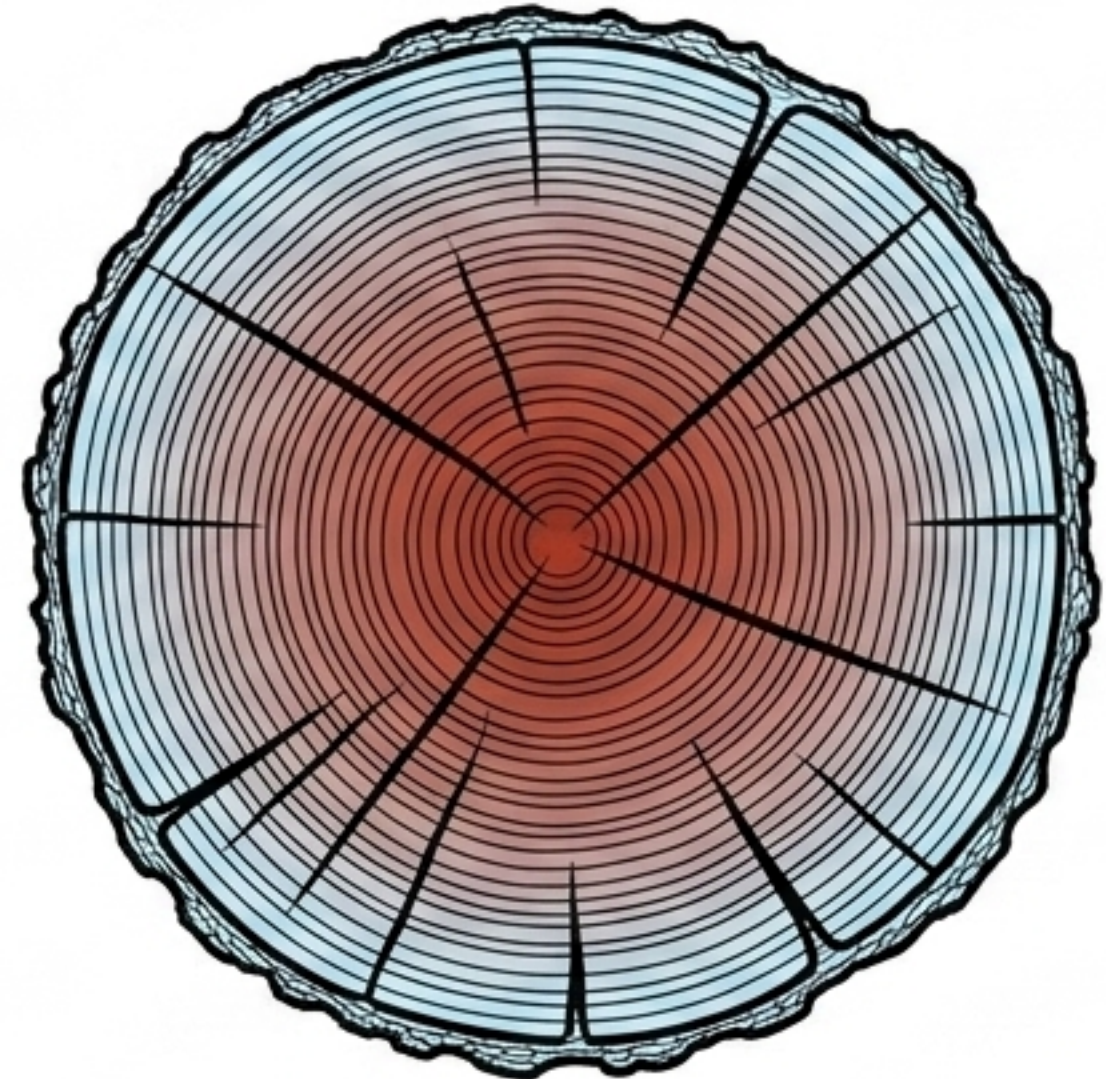
THE EVOLVING THERMAL GRID

Xylomorphic Computation: When substrate, process, and function are a single material fact. Derived from Greek xylon (wood) and morphē (form). Computation that cannot be pried apart from its physical function.

SUBSTRATE-INDEPENDENT

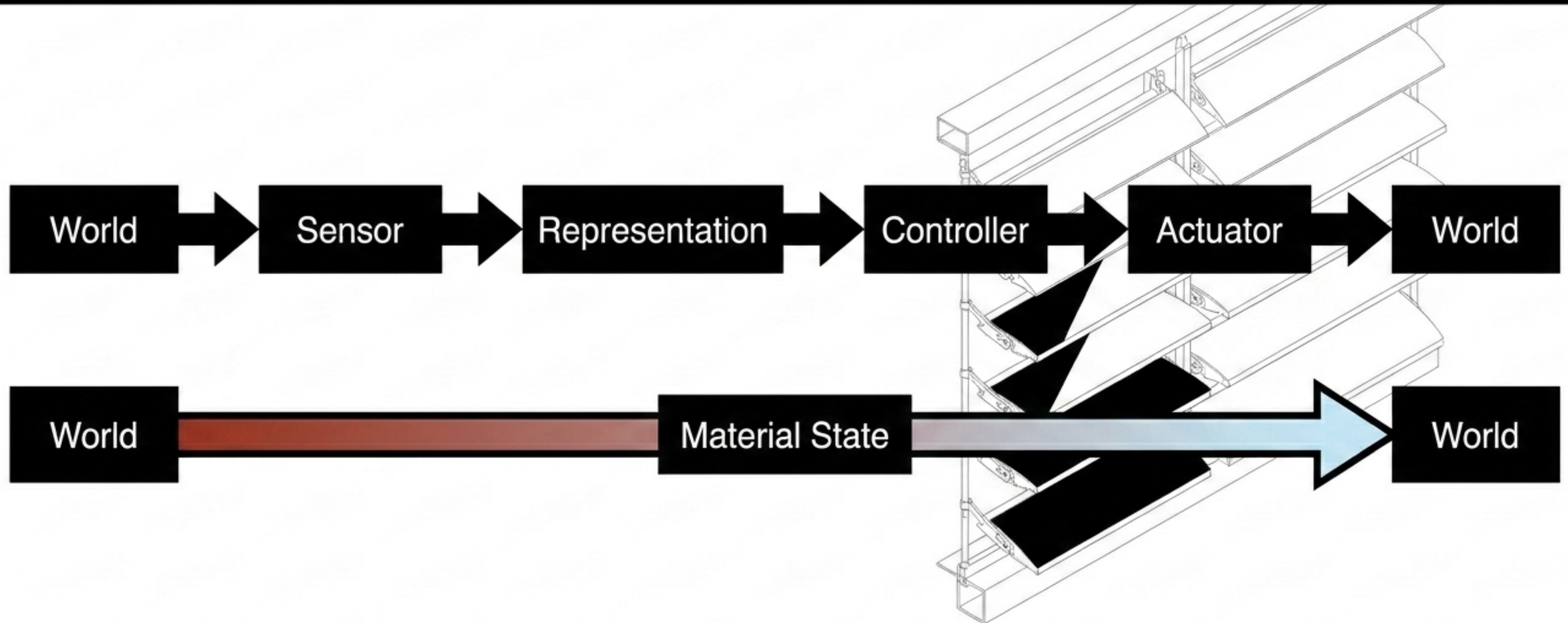


SUBSTRATE-FUSED



THE EVOLVING THERMAL GRID

Collapsing the control loop: Sensing, memory, and actuation as one physical event.
The wax's melting point (18°C) is the thermometer. Its expansion is the actuator.
The control layer has provably disappeared.

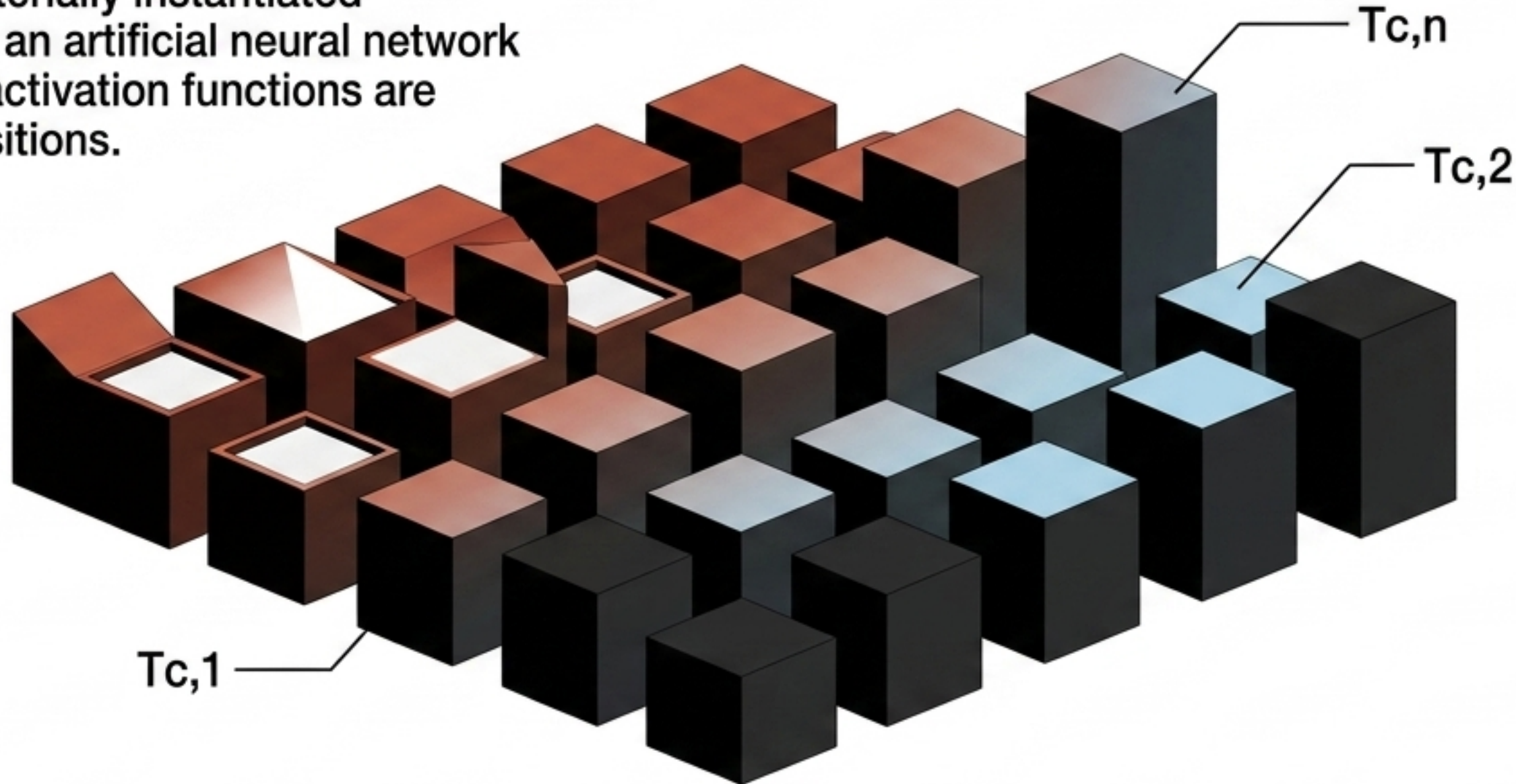


THE EVOLVING THERMAL GRID

Spatial Diversity: Translating continuous thermal space into computational thresholds

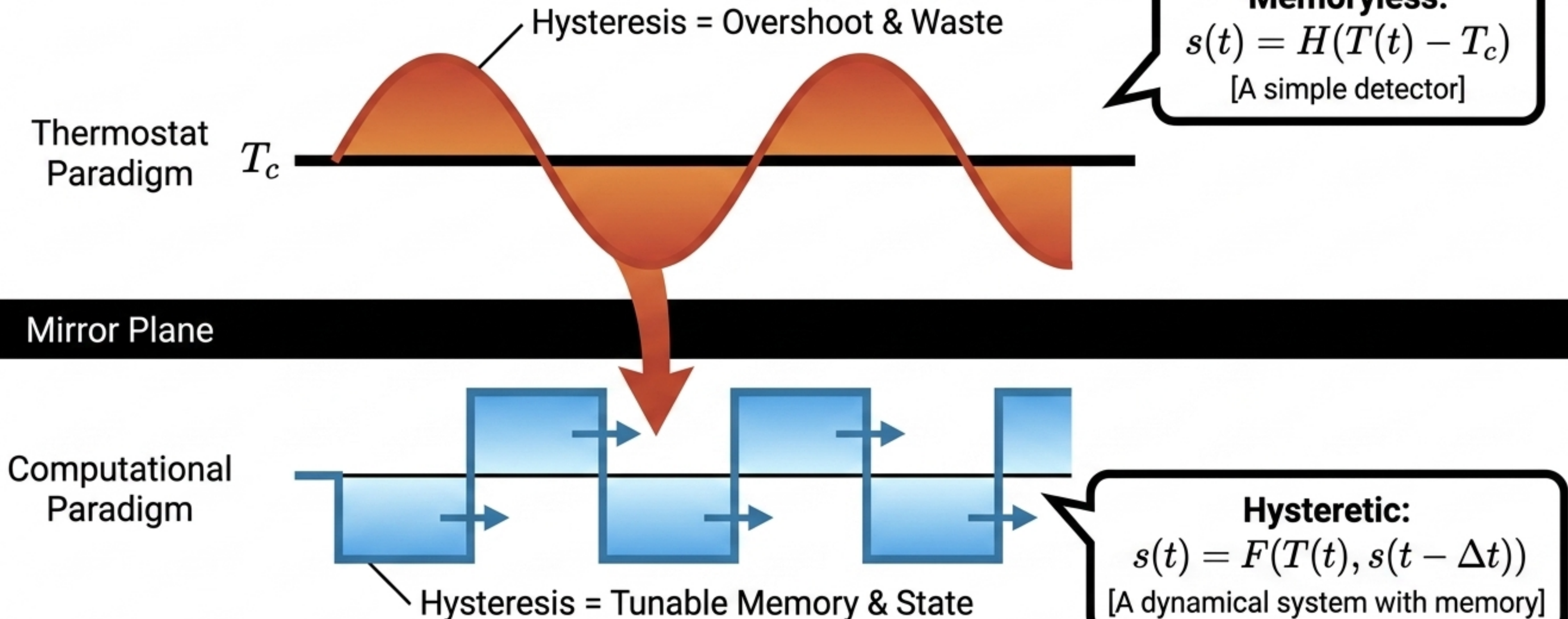
By deliberately varying the melting points of paraffin across an array, a uniform environmental input produces a complex spatial pattern of open and closed elements.

A heterogeneous, materially instantiated analogue of a layer in an artificial neural network—where the activation functions are physical melting transitions.



THE EVOLVING THERMAL GRID

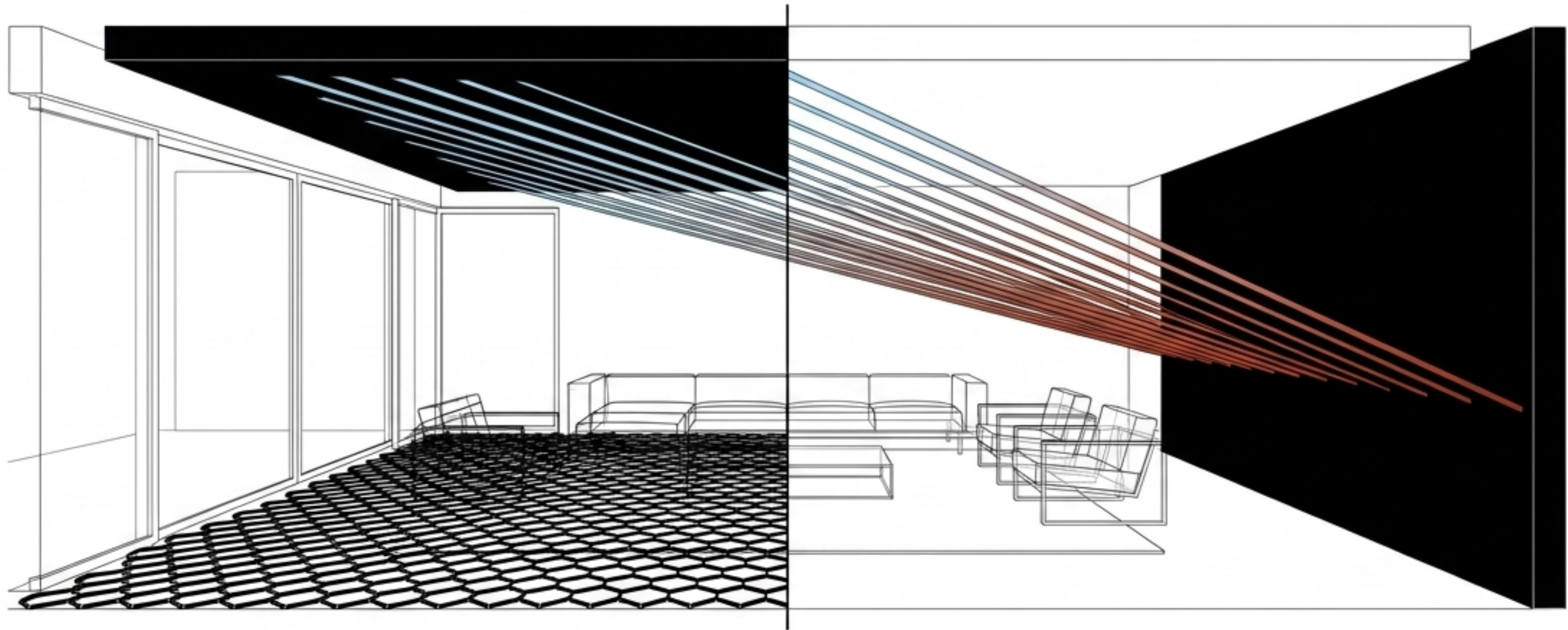
Hysteresis as Memory: Transforming a thermodynamic defect into a computational resource.



THE EVOLVING THERMAL GRID

The Coupling Continuum: Physical layout as the interaction matrix.

The coupling matrix isn't computed in a processor; it is physically inscribed into the room's sightlines and material spacing.



Local / Diffusive (Conduction)

Long-Range / Hopfield (Radiation)

THE EVOLVING THERMAL GRID

The Radiative Switch Field: A landscape of computational attractors built from wax and sightlines.

Exchanging infrared flux across open space is mathematically identical to a neural network descending an energy landscape toward a stable configuration.

1. Thresholds
(Varied melting points = Activation)

3. Coupling
(Sightlines/Conduction = Network Topology)

2. Hysteresis
(Thermal lag = Memory)

THE EVOLVING THERMAL GRID

Engineered vs. Emergent:

A river settles into a channel; a computer settles into a solution.

The constraint landscape is a byproduct of formation. Interpretable, but not intended.

Key Term:

Constraint Inscription.

The geometry, wax parameters, and thresholds are the program, written into matter rather than symbols.

The geometry, wax parameters, and thresholds are the program, written into matter rather than symbols.

The constraint landscape is inscribed in advance. Relaxing within it is informative, not incidental.

THE EVOLVING THERMAL GRID

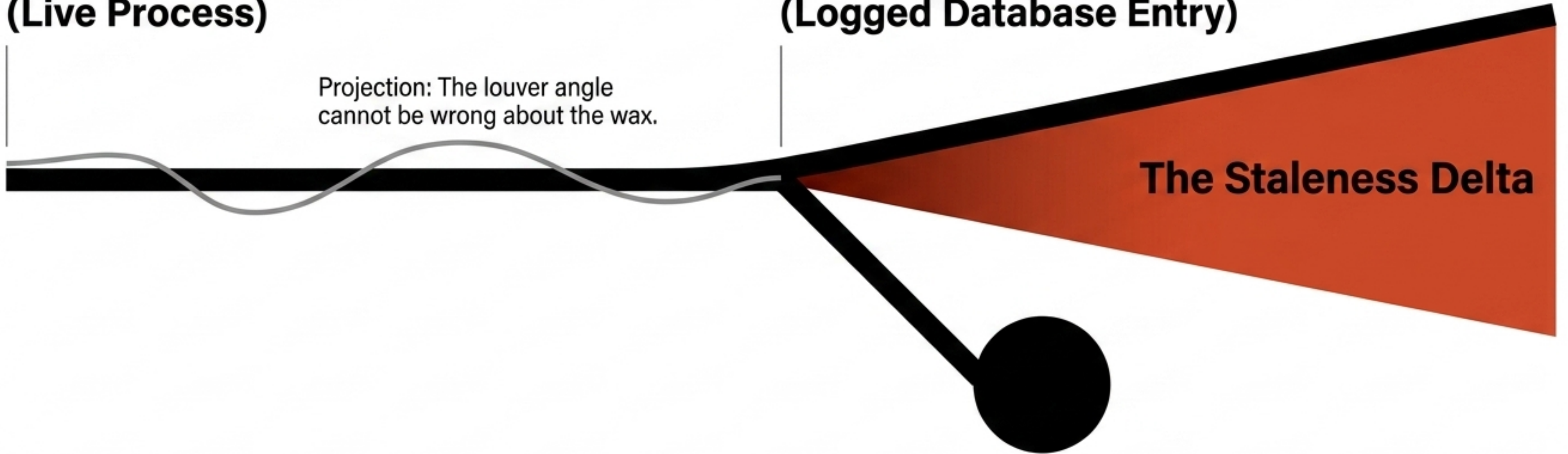
The Representation Trap: The capacity for a readout to become false.
Projection creates observability. Representation creates separability.

**Material State
(Live Process)**

**Representation
(Logged Database Entry)**

Projection: The louver angle
cannot be wrong about the wax.

The Staleness Delta

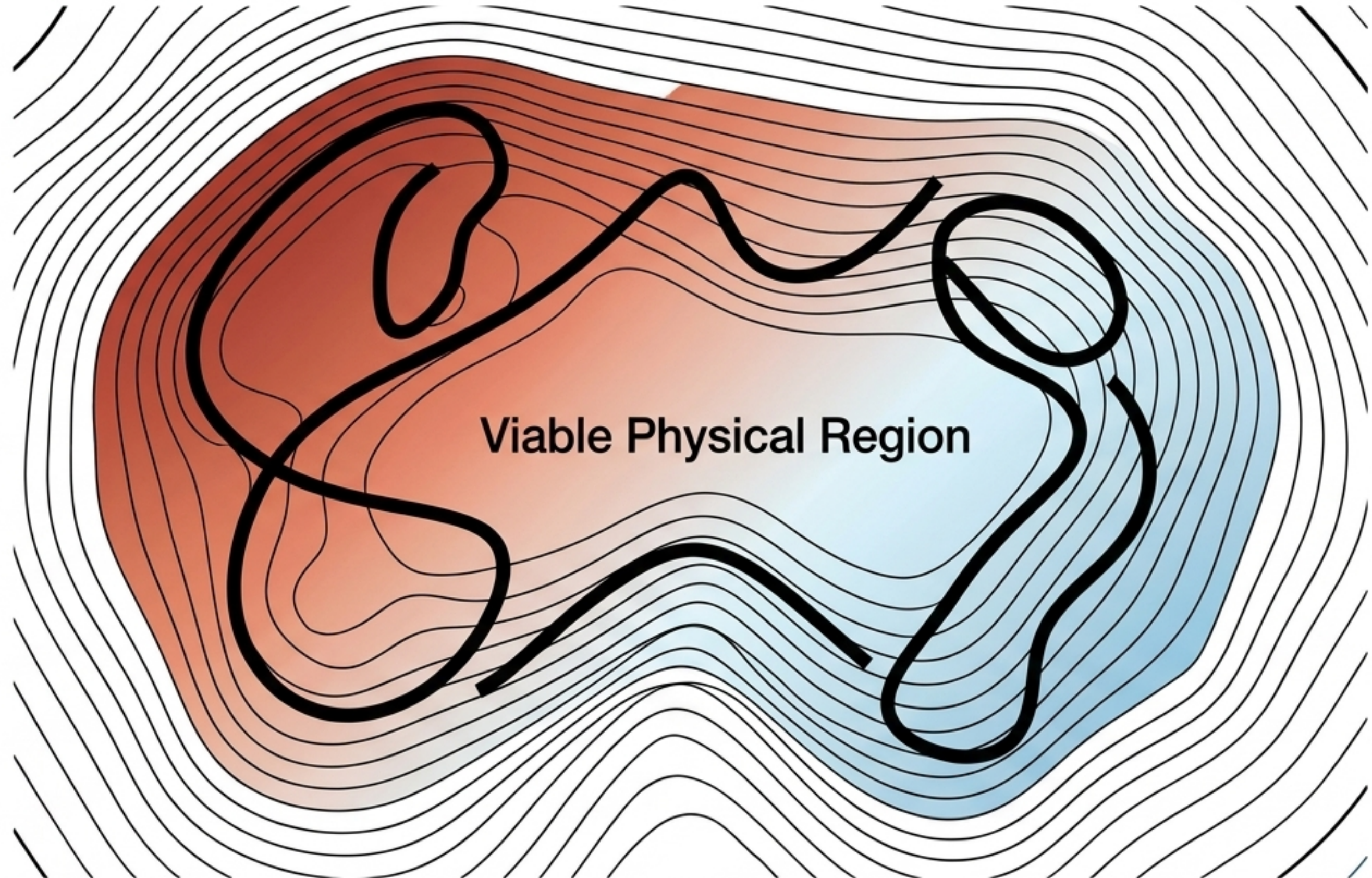


THE EVOLVING THERMAL GRID

Correctness without an answer key: Navigating viability regions.

A radiative switch field is correct not by hitting an externally specified target, but by remaining within the region specified as physically admissible when it was built.

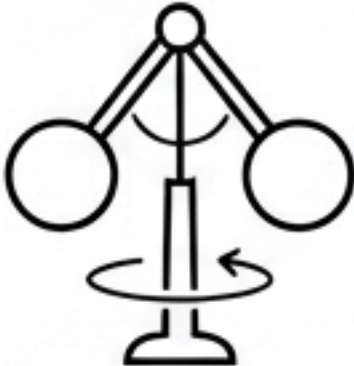
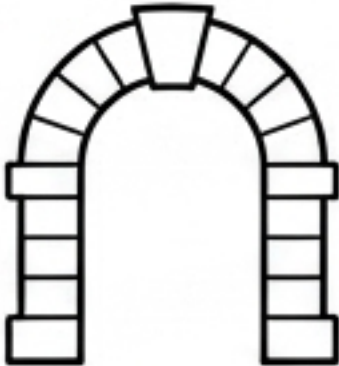
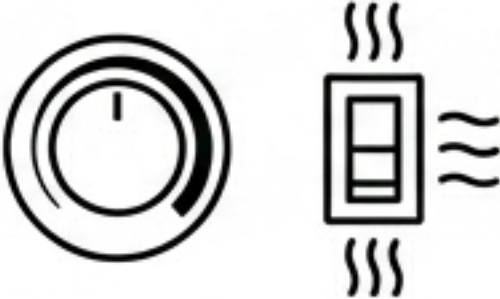
Self-maintenance replaces goal-seeking.



THE EVOLVING THERMAL GRID

Temporal Modes: The changing architecture of physical computation over time.

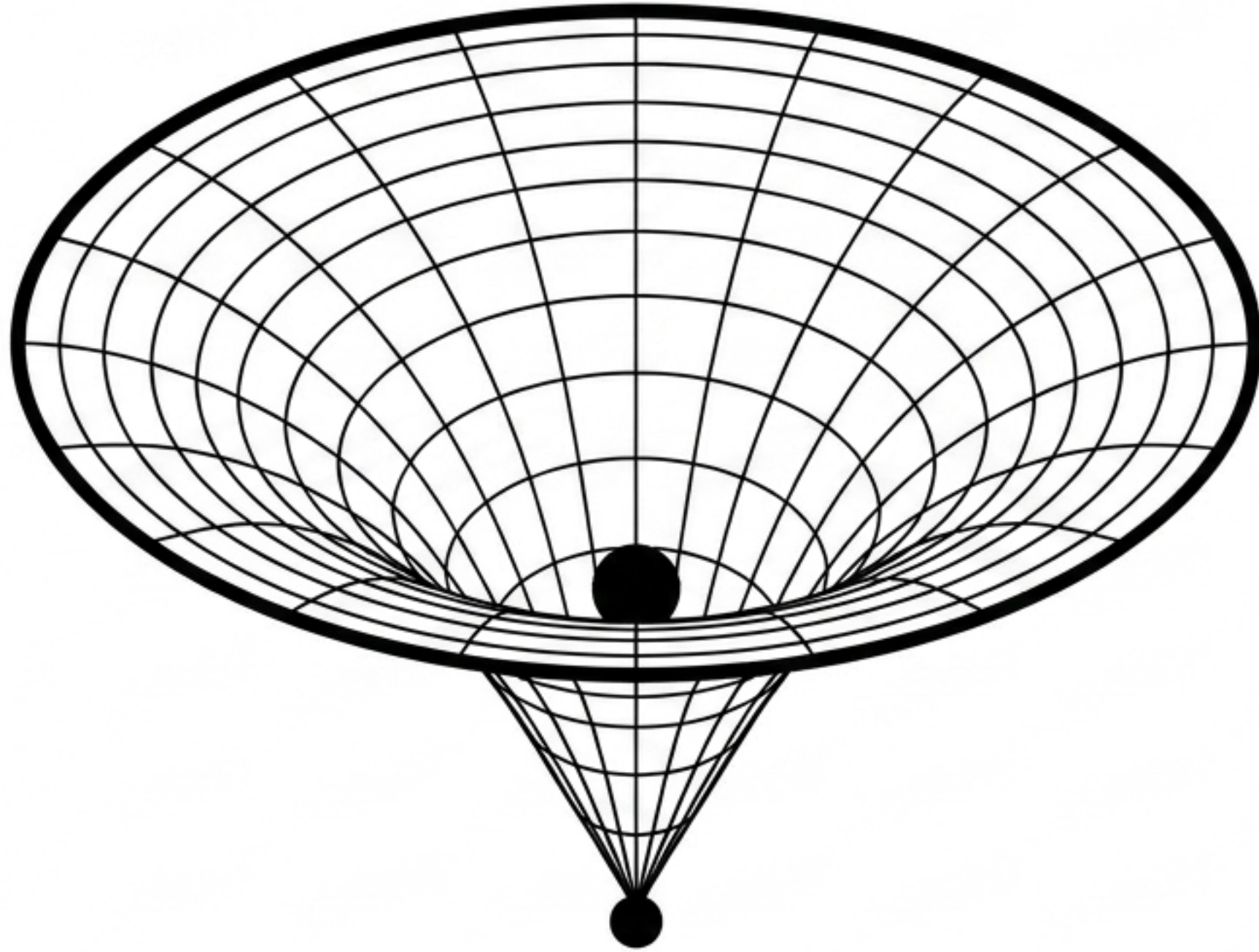
Computation is not always a fleeting process. Sometimes, the computation is over, and the structure left standing is the frozen trajectory of its logic.

	Continuous (exists while running)	Accumulative (layered history)	Terminal (frozen outcome)
Non-Thermal	Centrifugal Governor 	[Empty/N/A]	Settled Arch 
Thermal / Radiative	Thermostat / Active Switch Field 	Tree Rings / Stratified Residue 	Settled Field (under fixed constraints) 

THE EVOLVING THERMAL GRID

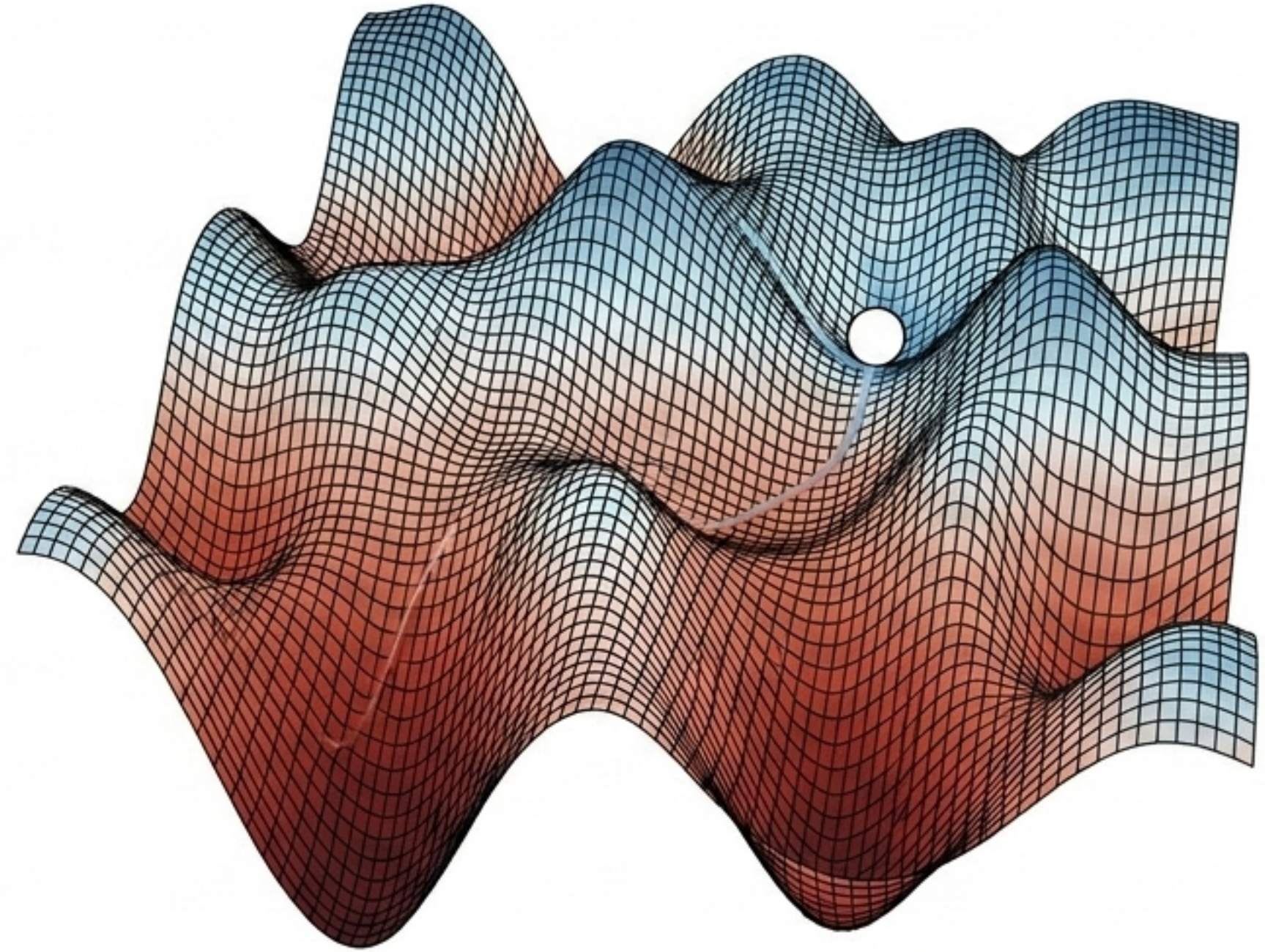
Constraint Richness: The difference between a stable heater and a true computer.

Closed but Degenerate:



Uniform thresholds. 1 attractor. No information carried by the final state.

Computationally Rich:



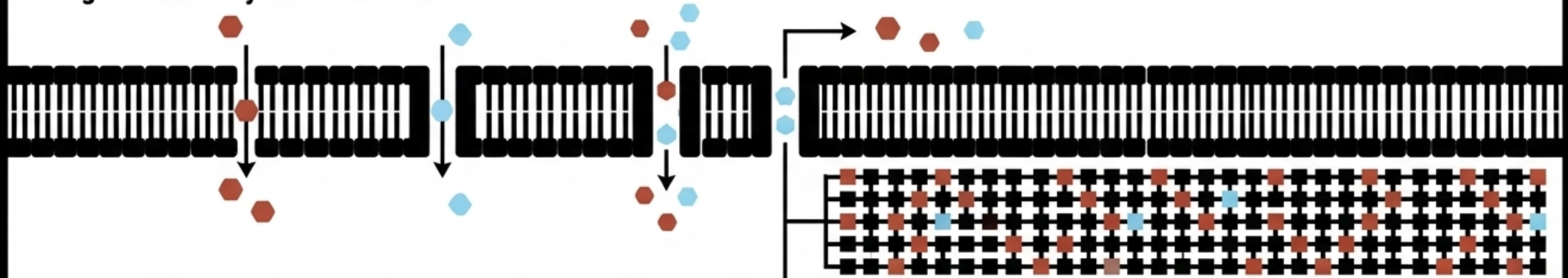
Heterogeneous thresholds. Many attractors. The settled state contains the history of the room's constraints.

THE EVOLVING THERMAL GRID

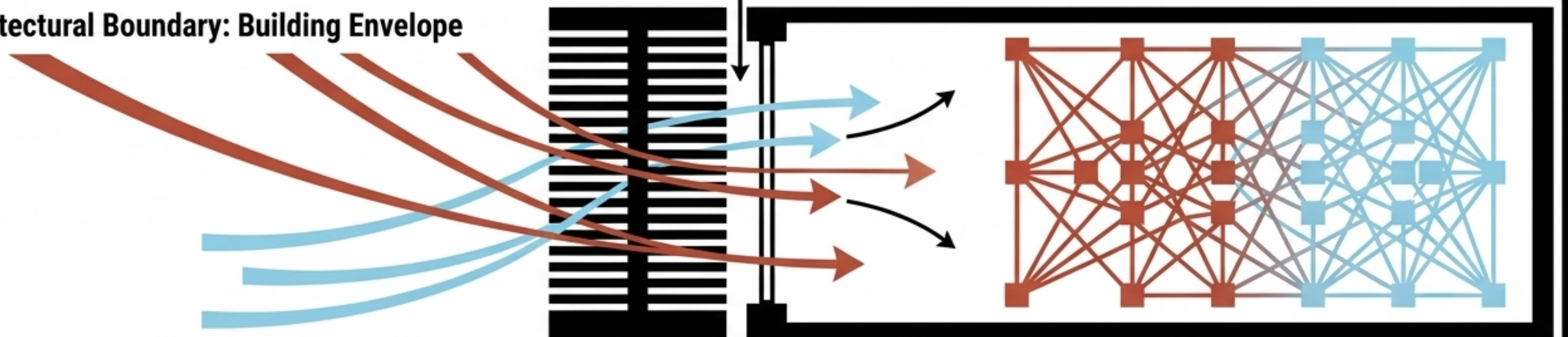
Permeability and Boundary: Containment is organization, not isolation.

Exchanging heat and moisture with the weather does not dissolve the computer. The boundary is the region where the field's internal coupling remains dominant over outside influence.

Biological Boundary: Cell Membrane



Architectural Boundary: Building Envelope



COMPUTATION AS CONTAINMENT

A conventional infrastructure object has an internal machine and an external world.
Xylomorphic computation collapses this distinction. The house does not host a computer.
While remaining a livable enclosure, it is one.

